

Predictive latent dynamics for parametric vortex-gust airfoil interactions at $\text{Re} = 5000$

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Abstract

We ask whether a predictive latent training objective yields a reduced-order state for parametric vortex-gust airfoil interactions that is more useful for forward-time forecasting than a reconstructive one. The data are numerical simulations of a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $\text{Re} = 5000$, perturbed by a discrete Taylor vortex parametrised by gust strength G , core diameter D , and wall-normal offset Y , spanning 84 cases partitioned into 226 training, 28 validation, 42 in-distribution test, and 24 out-of-distribution extrapolation encounters. Three encoder families are compared at matched latent dimension: a joint-embedding predictive architecture (JEPA), an observable-augmented reconstructive autoencoder in the lineage of Fukami & Taira, and a proper orthogonal decomposition (POD) basis. A single autoregressive transformer predictor, training recipe, parametric conditioning on (G, D, Y) , and probe family are held fixed across all three, so that the encoder is the only thing that varies. We judge each reduced state not by snapshot reconstruction but by *forward physical closure*: whether the predictor rollout keeps six physical observables, namely the lift and drag coefficients, the wall-normal impulse, the wake enstrophy, and the positive and negative circulation, close to the simulation out to a horizon of $H = 16$ frames after gust impact. On held-out encounters the predictive latent reduces the wake-enstrophy error by a factor of 2.4 to 3.0 relative to the reconstructive and linear baselines, with smaller but consistent gains on the force observables, while the linear basis remains competitive on the integrated impulse. The mechanism is a latent-drift diagnostic: under autoregressive rollout the reconstructive latent leaves its training manifold by an order of magnitude at $H = 16$, whereas the predictive and linear latents remain within it, so that the wake observables that are hidden from a reconstruction objective become recoverable from a predictive one. A conditioning-only floor shows that the held-out advantage is not explained by the parametric conditioning the predictor shares across baselines. Because the predictive encoder here also differs from the reconstructive baseline in architecture and in auxiliary observable supervision, we attribute the gain to the predictive latent family rather than to the training objective in isolation, and we set out the matched-architecture controls that would isolate it. The predictor is the latent dynamics function required by model-based control, and we discuss the pathway to closed-loop deployment and its current limits.

1 Introduction

Small uncrewed and micro air vehicles increasingly operate in airspace where the gust velocity is comparable to or larger than their flight speed, such as urban canyons, mountainous terrain, and the wakes of buildings and ships [9, 13]. In this regime the gust ratio G , the ratio of the characteristic gust velocity to the free-stream velocity, exceeds unity, and the interaction between a discrete vortex gust and a lifting surface produces large, fast transients in the aerodynamic loads that conventional gust models do not capture [5, 6]. The response is governed by the timing of the vortex impact relative to the airfoil’s own shedding, by the wall-normal offset of the vortex core,

and by the gust strength and core size. Even at the moderate chord Reynolds number $Re = 5000$, the post-stall flow at $\alpha = 14^\circ$ is massively separated, and the gust-induced disturbance interacts nonlinearly with the natural shedding cycle [6]. Resolving the full parametric envelope by simulation is expensive, so reduced-order models (ROMs) of the encounter-to-encounter and case-to-case variability are an active line of work, and a reduced model is also the object a controller would plan against.

Recent ROMs for separated and extreme-aerodynamic flows share a common ingredient: a reconstruction-trained autoencoder whose bottleneck is shaped by an auxiliary aerodynamic observable. Fukami & Taira [7] trained a convolutional autoencoder jointly with a lift-prediction head and found that the auxiliary loss reduced the intrinsic dimension of the discovered manifold for extreme vortex-gust flows from roughly ten to three. The same observable-augmentation idea was extended across multi-source turbulent data [8], and was used by Solera-Rico et al. [18] with a β -variational autoencoder and a transformer ROM on a vortex-gust airfoil dataset of the same family considered here. Across this lineage the reconstruction term anchors the latent geometry to the instantaneous flow field, and an observable-prediction term orients it along aerodynamically relevant directions. A recurring theme of the geometric-analysis literature, however, is that the latent coordinates of an autoencoder are unique only up to a diffeomorphism, so reconstruction leaves the geometry of the latent largely unconstrained, which complicates any downstream task that relies on that geometry, including dynamics, interpolation, and comparison [19, 10].

A complementary route to a reduced state is to drop the reconstruction term altogether and to train the encoder and a predictor end to end on latent-space forecasting, with an explicit anti-collapse regulariser in place of the decoder. This is the joint-embedding predictive architecture (JEPA) [11]: an encoder maps the input to a representation, a predictor advances that representation in time, and the loss never reconstructs the field. Two consequences follow. The encoder is free to discard detail that is not predictive of the future, and the predictor cannot exploit pixel-level shortcuts because no field-space signal reaches it. The image and video instantiations [1, 2] and the recent from-pixels variants that remove the moving-target encoder in favour of a characteristic-function regulariser [12, 3] or a variance-covariance objective [17, 4] have so far been evaluated on gridworld and toy visual domains. Their behaviour on a low-intrinsic-dimension physics dataset, where the data sit on a roughly five- to ten-dimensional manifold, is uncharacterised.

The question this paper addresses is therefore not which encoder reconstructs the flow best, but which produces a reduced state that remains useful when propagated forward in time. We treat this as a fluid-mechanics question with a physical answer: a reduced state for gust-airfoil forecasting should be judged by whether its forward evolution keeps physically meaningful observables close to the simulation, and by whether the latent trajectory it generates stays on the manifold the model was trained on. We make three contributions.

First, we set up a controlled comparison in which a predictive encoder (JEPA), a reconstructive encoder in the Fukami lineage, and a POD basis are evaluated at matched latent dimension under a single shared autoregressive predictor, training recipe, parametric conditioning on $\mathbf{c} = (G, D, Y)$, and probe family, so that forecast quality is attributable to the encoder alone (§3). The figure of merit is forward physical closure of the rollout to six observables at horizon $H = 16$ (§4.1). On held-out encounters the predictive latent closes the spatially distributed wake observables markedly better than the reconstructive and linear baselines, while the linear basis remains competitive on the integrated impulse.

Second, we identify the mechanism. A latent-drift diagnostic shows that the reconstructive rollout leaves its training manifold by an order of magnitude at $H = 16$, whereas the predictive and linear rollouts remain within it (§4.3). We strengthen this from a single distance into a geometric and topological statement: the predictive latent organises the encounter as a closed cycle whose topology survives rollout, and its latent metric tracks the physical transport geometry

of the flow more faithfully than the reconstructive latent does.

Third, we are explicit about what the comparison does and does not isolate. The predictive encoder differs from the reconstructive baseline not only in its objective but also in its architecture and in its auxiliary observable supervision. We therefore phrase the result as a property of the *predictive latent family*, report a conditioning-only floor that rules out the shared parametric conditioning as the explanation (§4.1), and set out the matched-architecture and matched-auxiliary controls required to attribute the gain to the predictive objective in isolation (§4.5). We close by relating the predictor to the latent dynamics function of model-based control, including a deployment pilot that does not yet meet its tolerances and is reported as a limitation rather than a result (§5).

2 Flow configuration and data

2.1 Vortex gust-airfoil interaction

We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $Re = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The governing data are the spanwise-periodic simulations of Fukami, Smith & Taira [6], in which the same configuration was characterised in detail. The gust is parametrised by three scalars: the signed gust ratio G , the core diameter D , and the wall-normal offset Y/c of the vortex centre relative to the leading edge. The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y/c \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages [16, 6, 15]. The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The six observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation [6] is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

2.2 Data, fields, and observables

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05 t/c$, one encounter per gust release. The dataset comprises 84 cases partitioned into training, validation, an in-distribution test set (test_b), and an out-of-distribution test set (test_c). The in-distribution test set is stratified to span the five gust strengths, the three core diameters, and the two source groups; test_c is the $|G| = 4$ extrapolation set. At the horizon

$H = 16$ used throughout, the partition contains 226 training, 28 validation, 42 test_b, and 24 test_c encounters. The undisturbed reference case is retained in training. The partition is fixed once and used identically by every model and baseline.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\text{wall}}(192)$, and the lift and drag coefficients C_L, C_D . The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by the training standard deviation with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures.

We evaluate forward closure against six physical observables that together span body-force, integrated-flow, and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wall-normal impulse I_y is an integrated-flow quantity coupled to the wake-vortex dynamics. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature.

3 Predictive latent representation and evaluation protocol

3.1 Encoder, predictor, and the conditioning split

The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at $d \in \{32, 64\}$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is *unconditional* by design: it sees only the flow field, not the gust parameters $\mathbf{c} = (G, D, Y)$. This is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

The predictor P_ϕ advances the latent in time. It is a six-layer autoregressive transformer with rotary positional embeddings and a causal mask, conditioned on $\mathbf{c}$ through adaptive layer-norm modulation at every block; its internal structure is given in Appendix A. The predictor is the only stage that sees $\mathbf{c}$. Figure 1 shows the encoder and predictor, and Figure 2 contrasts the predictive training signal with the reconstructive one: the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error, whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss.

3.2 Training objective and the auxiliary observable heads

The predictive encoder and predictor are trained jointly on a one-step latent-prediction term, a scheduled-sampling rollout term that exposes the predictor to its own errors, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of [3, 12], which matches the latent distribution to an isotropic Gaussian through a small number of random projections; we found the choice of anti-collapse regulariser to be second order relative to a variance-covariance alternative [4] at this data scale, and we report that comparison in Appendix A.

Two auxiliary heads read aerodynamic observables off the encoder output: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, and a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, trained with small weights jointly with the representation. These heads are adopted from the observable-augmentation lineage [7, 8]. They are also, however, the reason the present predictive encoder is not a clean isolation of the training objective: the reconstructive baseline below carries a lift head but not a wake head, and the two encoders differ in architecture as well. We are explicit about this throughout. The headline comparison is therefore between encoder *families* as configured, and the controls of §4.5 are designed to separate the objective from the architecture and the wake supervision.

3.3 The matched-predictor fairness protocol

The comparison rests on a single principle: every baseline is the same three-stage pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields; an identical autoregressive transformer predictor rolls it forward one step at a time given $\mathbf{c}$; and an identical probe family maps the predicted latent to a physical observable for evaluation (Figure 3). The predictor architecture, its training recipe, the parametric conditioning, and the probe family are held fixed across the three encoder families. The latent dimension is matched within each family: JEPA at $d \in \{32, 64\}$, the reconstructive autoencoder at $d \in \{3, 32, 64\}$ to include both the published $d = 3$ recipe and matched-capacity variants, and POD at $d \in \{16, 32, 64\}$. This isolation is the only fair way to attribute forecast quality to the representation rather than to incidental differences in the predictor or its training.

3.4 Forward physical closure

The figure of merit is forward physical closure. Given a ground-truth $\mathbf{c}$ and an initial latent, the predictor is rolled recursively to $H = 16$, and at each step the probe maps the predicted latent to each observable. A small closure error means the encoder produces a latent whose dynamics, propagated by the shared predictor, retain the information the probe needs to recover the physical quantity; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. We report closure on the held-out `test_b` and `test_c` encounters as the mean absolute error against the simulation, with bootstrap 95% confidence intervals over encounters and the coefficient of determination R^2 . **[PENDING: compute held-out `test_b`/`test_c` R^2 for all baselines and populate Table 3.]**

3.5 Two mechanistic diagnostics

Latent drift. A predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H , measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

Conditioning-only floor. Because the predictor is conditioned on $\mathbf{c} = (G, D, Y)$, a natural concern is that the forecast quality is carried by the conditioning rather than by the latent. We bound this with a kernel-ridge regressor that maps $\mathbf{c}$ directly to each observable at the impact frame, with no latent input, fit on training and evaluated on the held-out splits. This is the floor any learned encoder must clear on held-out cases to demonstrate that its latent contributes beyond what is already implicit in the parameters.

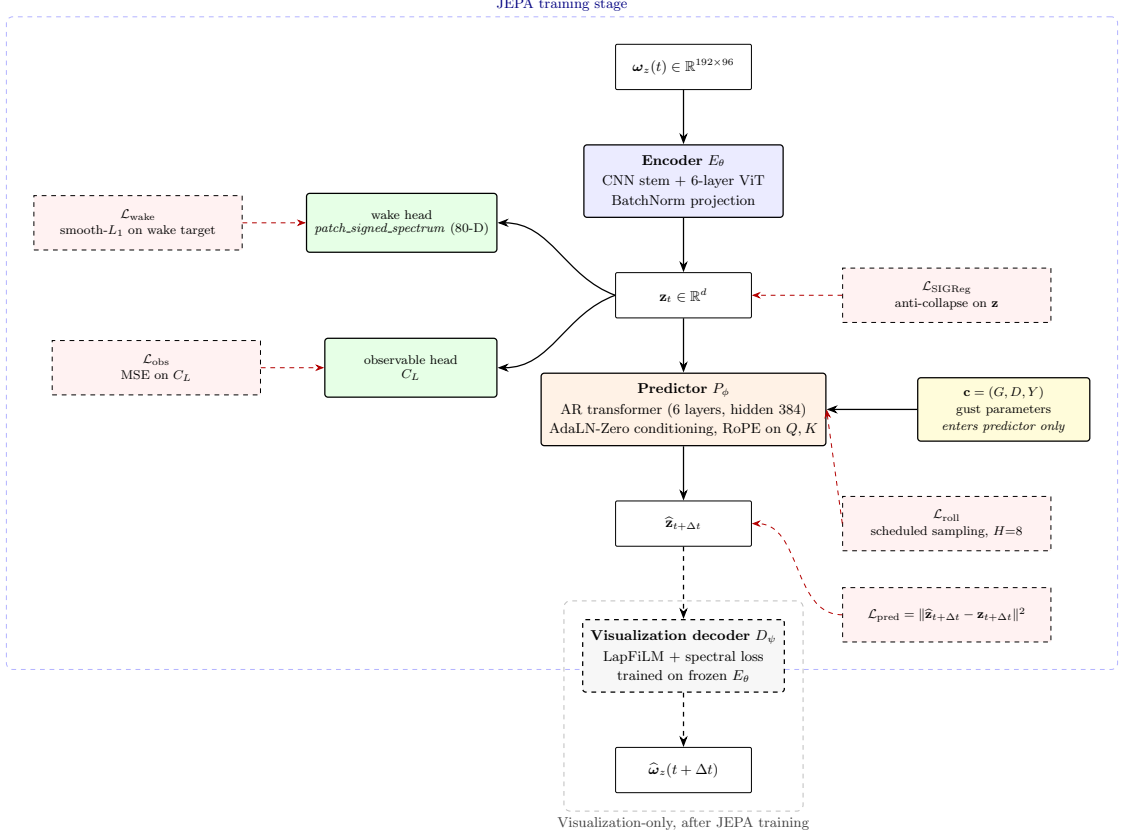


Figure 1: Predictive encoder and predictor. The encoder is unconditional and maps the mid-plane vorticity field to a latent $\mathbf{z}_t$. The autoregressive transformer predictor advances the latent given the gust conditioning $\mathbf{c} = (G, D, Y)$. The two auxiliary observable heads (lift and wake) and the visualisation decoder are attached to the frozen encoder and are never part of the latent-prediction loss.

4 Results

4.1 Forward physical closure across encoder families

Table 1 summarises the three encoder families and the protocol. The held-out closure is reported in Table 2 as the mean absolute error of the $H = 16$ rollout against each observable on the test_b encounters, and graphically in Figure 4 across test_b and test_c. The predictive latent gives the lowest held-out error on the wake observables by a wide margin: at $d = 64$ its wake-entrrophy error is 29.83 against 72.90 for the reconstructive autoencoder and 89.31 for the linear basis, a factor of 2.4 and 3.0 respectively, and its positive and negative circulation errors are likewise the smallest. The gap narrows on the scalar forces, which all three families approximate more evenly, although the predictive latent still leads on C_L (0.624 against 0.989 and 0.872). The integrated impulse I_y is where the linear basis is most competitive: on the in-distribution training fit (Table 4) POD attains the highest I_y score of any family, an expected consequence of its energy-optimal projection, although on held-out error the predictive latent still leads there. We flag this train-versus-held-out reversal because it is exactly why the held-out coefficient of determination, not the training fit, must head the comparison.

That held-out coefficient of determination is the one number the original analysis did not report; it reported a training-set R^2 (Table 4, mean 0.835 at $d = 64$), which measures in-distribution fit rather than generalisation. The held-out R^2 follows directly from the same rollout

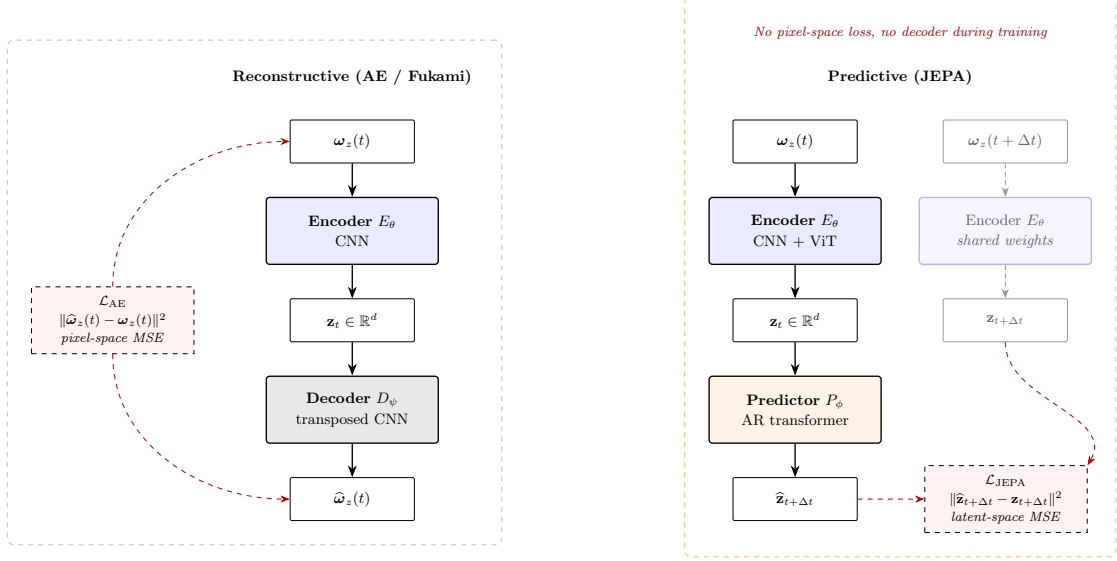


Figure 2: Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t , decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training.

Table 1: Encoder families compared under the matched-predictor protocol of §3.3. All three share one autoregressive transformer predictor, training recipe, conditioning on $\mathbf{c} = (G, D, Y)$, and probe family; only the encoder differs. “Aux. heads” lists the observable supervision attached to the encoder. The architecture and auxiliary-head columns differ across families, which is why the headline result is attributed to the predictive *family* and isolated only by the controls of §4.5.

Family	Objective	Encoder	Latent dim. d	Aux. heads
JEPA (this work)	predictive (latent)	CNN + ViT	32, 64	lift + wake
Reconstructive AE	reconstructive (field)	CNN	3, 32, 64	lift
POD	linear projection	—	16, 32, 64	—

predictions already used for Table 2 and belongs in Table 3. **[PENDING: populate Table 3 and rewrite this paragraph’s summary statistic in held-out R^2 .]**

The conditioning-only floor (Table 6) confirms that the held-out result is not an artefact of the parametric conditioning the predictor shares. A kernel-ridge map from $\mathbf{c} = (G, D, Y)$ to the impact-frame observables interpolates the training points well, as any flexible kernel in a three-dimensional input will, but collapses on test_b and goes negative on test_c for five of six observables. The parameters alone cannot generalise outside the training envelope, so the held-out closure is doing representational work that the conditioning cannot replicate.

4.2 Horizon dependence

A dynamics paper should not rest on a single horizon. Comparing recursive (Markov) rollout against a one-shot prediction at horizon H , the two agree to within the encounter-level scatter for $H \leq 16$ and diverge for $H \geq 32$ as the recursive error compounds, which places the reported $H = 16$ inside the regime where the rollout mode does not affect the conclusion and which matches the planning horizon relevant to the impact-instant load. The full decay of closure with horizon, swept over $H \in \{1, 4, 8, 16, 32, 64\}$ for every family on test_b and test_c, is the natural extension. **[PENDING: run the horizon sweep and add the closure-versus- H figure.]**

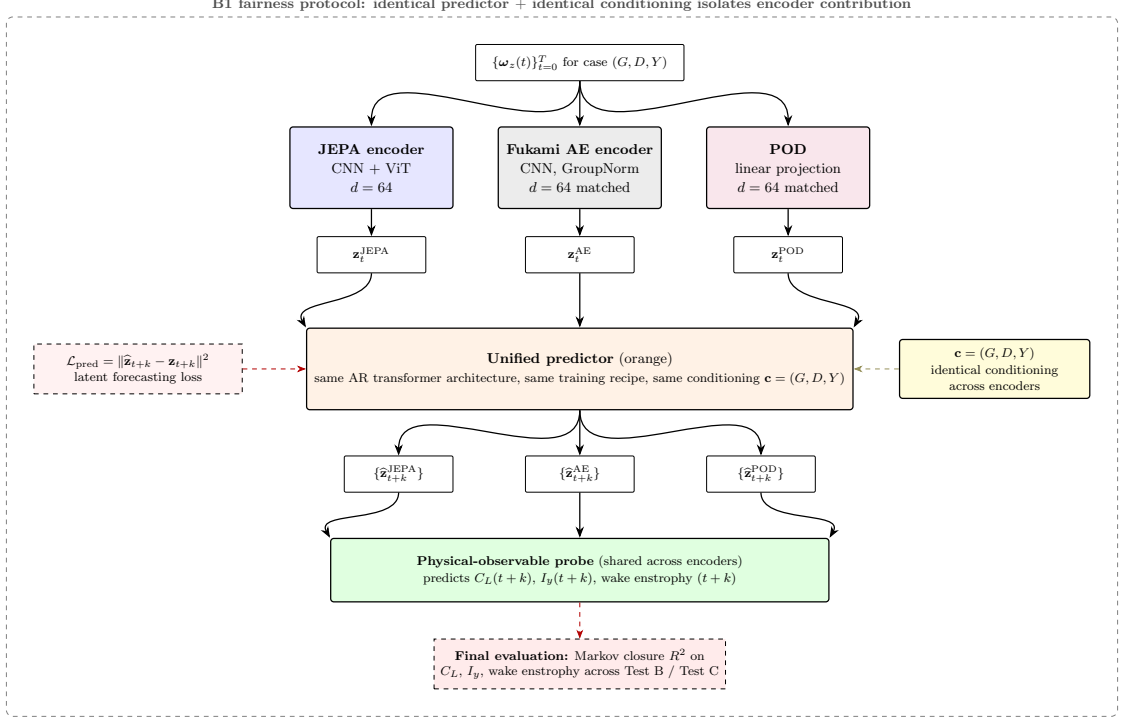


Figure 3: Matched-predictor fairness protocol. Each encoder family produces its own latent stream, which is then advanced by an identical autoregressive predictor with identical conditioning $\mathbf{c} = (G, D, Y)$ and read by an identical probe family. Closure quality is attributable to the encoder alone.

4.3 The drift mechanism: distance, topology, and transport geometry

The closure gap is large enough that its mechanism deserves direct measurement.

Distance. The latent-drift diagnostic (Table 5) measures how far each rollout leaves its training manifold. After $H = 16$ steps the reconstructive rollout sits an order of magnitude further from its training distribution than its simulation-encoded counterpart (ratio 9.90), while the predictive and linear rollouts remain within it (ratios 0.85 and 0.81). This is the mechanism behind the closure gap: the probe attached to a drifted rollout is queried outside the support it was fit on, so its output degrades for reasons unrelated to the predictor’s one-step quality. A reconstruction objective places latent codes wherever the decoder finds convenient and does not constrain the geometry to be predictable in time; the predictive objective, regularised against collapse, produces a latent that supports stable rollout by construction. The pre-impact inference of the impact lift makes the same point from the encoder side: an oracle probe on the simulation-encoded predictive latent recovers the impact C_L at $R^2 = 0.683$ at every lead time, whereas the same oracle on the reconstructive latent is negative (-0.185), a representation failure rather than a probe failure, since no probe can recover information the latent does not carry. Rolling the predictive latent forward through the predictor recovers about 0.60 of that skill from five frames before impact and about 0.35 from ten, several frames of usable forecast horizon beyond a direct wall-pressure read at the same lead times.

Topology. A single distance understates the geometric content of the result. A gust encounter is, in state space, a closed cycle: the flow departs the baseline limit cycle, executes the LEV-formation and shedding excursion, and returns [16]. We characterise this with persistent

Table 2: Held-out forward closure. Mean absolute error at horizon $H = 16$ on the $n = 42$ test_b encounters, for the rollout from the simulation-encoded initial latent, against six physical observables. Smaller is better; best per column in bold. The wake-entrophy and circulation columns are in native units, the force coefficients are dimensionless. This held-out error, not the training fit of Table 4, is the headline evidence.

Baseline	d	C_L	C_D	I_y	wake-entst.	circ. pos.	circ. neg.
JEPA	64	0.624	0.301	1.901	29.83	0.801	0.715
JEPA	32	0.675	0.246	2.976	37.99	1.084	1.061
Fukami	3	1.074	0.384	2.327	51.70	1.110	1.886
Fukami	32	1.038	0.368	2.195	63.94	0.946	1.789
Fukami	64	0.989	0.340	2.421	72.90	1.057	1.904
POD	16	1.168	0.317	2.235	89.14	1.645	1.681
POD	32	0.937	0.269	2.608	81.82	1.650	1.617
POD	64	0.872	0.291	2.245	89.31	1.660	1.800

Table 3: Held-out forward closure, coefficient of determination R^2 at $H = 16$ on test_b (and test_c), to be computed from the same rollout predictions used for Table 2. This table replaces the training-set R^2 of Table 4 as the reported closure quality.

Baseline	d	C_L	C_D	I_y	wake-entst.	circ. pos.	circ. neg.	mean
[PENDING: generate held-out R^2 from the rollout predictions]								

homology, which identifies the persistent one-dimensional cycle (H_1 generator) of a trajectory by a Vietoris–Rips filtration [16], applied here to four latent point clouds per encounter: the simulation-encoded predictive latent, its rollout, the simulation-encoded reconstructive latent, and its rollout. The prediction this makes, and the figure it produces, is that the predictive rollout preserves the encounter’s persistent H_1 generator (the cycle survives) while the reconstructive rollout’s generator collapses or fragments as the trajectory leaves the manifold, turning the scalar drift ratio into a coordinate-free topological signature. **[PENDING: compute persistence diagrams of the four latent trajectories and add the H_1 -lifetime-versus-horizon panel.]**

Transport geometry. The geometric reason the predictive rollout stays on the manifold can be made physical. Euclidean distance between vorticity fields is insensitive to the spatial transport of structures, whereas an unbalanced optimal-transport (OT) distance measures the cost of rearranging one field into another and so tracks advection of the LEV and shear layer [19]. Computing the OT distance between simulation fields along an encounter gives a physical transport geometry against which the latent metric can be checked: the prediction is that distances in the predictive latent track the OT geodesic more faithfully than distances in the reconstructive latent, so that a single predictor step corresponds to a physically smooth, transport- consistent change and iterating it does not leave the data manifold. **[PENDING: compute the OT-geodesic alignment of each latent metric and report the correlation, predictive versus reconstructive.]**

4.4 Where closure holds and fails: parameter and phase dependence

Which gust axes the latent resolves. Probing the predictive latent for the gust parameters on test_b recovers the gust strength G and the core diameter D ($R^2 = 0.46$ and 0.80) but barely resolves the wall-normal offset Y ($R^2 = 0.10$). This is a property of the data rather than of

Table 4: Training-set fit, for reference only. Coefficient of determination R^2 for predicting the six observables from the rolled-out latent ($H = 16$) evaluated *on the training distribution*. This quantifies in-distribution fit and is *not* the headline; held-out closure is in Tables 2–3. Best per column in bold; row mean at right.

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	0.864	0.802	0.562	0.934	0.922	0.927	0.835
JEPA	32	0.863	0.795	0.490	0.898	0.901	0.897	0.808
Fukami	3	0.553	0.408	0.191	0.079	0.130	0.033	0.232
Fukami	32	0.695	0.596	0.278	0.333	0.369	0.313	0.430
Fukami	64	0.671	0.589	0.335	0.277	0.336	0.352	0.427
POD	16	0.763	0.720	0.514	0.574	0.582	0.527	0.613
POD	32	0.763	0.748	0.690	0.589	0.618	0.621	0.671
POD	64	0.641	0.656	0.799	0.373	0.388	0.506	0.561

Table 5: Latent-drift diagnostic on test_b at $H = 16$. The Mahalanobis distance of the rolled-out latent is reported alongside that of the simulation-encoded latent in the same reference distribution; their ratio $r = d_{\text{Mahal, Markov}}/d_{\text{Mahal, DNS}}$ measures how far the autoregressive rollout has left the training manifold. A ratio near one means the rollout stays in distribution.

Baseline	d	$d_{\text{Mahal, ref}}$	$d_{\text{Mahal, rollout}}$	ratio r
Fukami	64	2.92	28.96	9.90
JEPA	64	8.08	6.84	0.85
JEPA	32	5.43	4.65	0.86
POD	64	7.74	6.27	0.81

the architecture: the training mass concentrates near $Y \approx 0$, so encounters at different offsets within the envelope are weakly separated along Y , and the latent inherits that weakness. Any deployment-time parameter estimate should treat Y as a nuisance variable the encoder resolves only weakly.

Phase relative to the baseline cycle. Because the undisturbed wake is a limit cycle (§2.1), an encounter is naturally described by its phase and amplitude relative to that cycle, the same reduction used to design gust-mitigation control for these flows [5]. Reading the predictive latent this way makes “recovery” precise: it is the return of the latent trajectory to the baseline limit cycle. The analysis we add is to verify that the predictive latent of the undisturbed case traces a closed periodic orbit, to define a phase along it, and to show that the predictor rollout carries the disturbed trajectory back toward that orbit for the predictive latent while the reconstructive rollout departs from it, which both defines recovery and connects the predictor to the phase-amplitude machinery of control. The persistent cycle of the previous subsection and this limit cycle are the same object viewed topologically and dynamically. **[PENDING: construct the latent limit cycle and phase, and show return-to-orbit under rollout, predictive versus reconstructive.]**

Where the error concentrates. The average closure hides where each family fails physically. Stratifying the held-out error by gust regime, namely the sign and magnitude of G , the diameter D , the offset Y , and the encounter phase (pre-impact, impact, peak load, recovery), would localise the predictive latent’s residual error and test whether the reconstructive failure concentrates in the wake-sensitive shedding and recovery phases, organised by physical effect as in the source characterisation of these flows [6]. **[PENDING: add the parameter- and phase-stratified**

Table 6: Conditioning-only floor. A kernel-ridge regressor maps the gust parameters $\mathbf{c} = (G, D, Y)$ directly to each observable at the impact frame, with no latent input; R^2 is reported on each split. This is the floor any learned encoder must clear on held-out cases to show its latent contributes beyond the parameters. The conditioning floor interpolates the training points but collapses on the held-out splits, going negative on test_c for five of six observables.

Observable	n_{train}	train R^2	test_b R^2	test_c R^2	JEPA $d=64$ test_b R^2
C_L	226	0.895	0.303	0.547	[PENDING:]
C_D	226	0.941	0.386	-2.459	[PENDING:]
I_y	226	0.422	-0.301	-1.335	[PENDING:]
wake-ensrophy	226	0.806	0.482	-1.252	[PENDING:]
circ. pos.	226	0.792	-0.032	-0.249	[PENDING:]
circ. neg.	226	0.799	0.568	-4.262	[PENDING:]

Table 7: Objective $\times$ architecture controls (decisive experiment). Mean held-out closure (R^2 at $H = 16$, test_b) and drift ratio for the four cells crossing {predictive, reconstructive} with {CNN, CNN+ViT} at $d = 64$, with the auxiliary heads matched across cells, plus a JEPA variant with the wake head removed. This table isolates the objective from the architecture and from the wake supervision.

Objective	Encoder	Aux. heads	mean closure R^2	drift ratio
[PENDING: run the 2×2 (auxiliary-matched) and the no-wake-head JEPA; populate]				

closure analysis.]

4.5 Isolating the cause: objective, architecture, and auxiliary-head controls

The results so far establish that the predictive latent *family*, as configured, closes the forward observables better than the reconstructive and linear families, and that the mechanism is reduced rollout drift. They do not yet establish that the *predictive objective in isolation* is responsible, because the predictive encoder differs from the reconstructive baseline in two further ways: its CNN+ViT architecture, and its 80-dimensional wake supervision. The latter is the more pointed confound, since the largest part of the reported advantage is on the wake observables that the predictive encoder alone was supervised on. Isolating the objective requires the controls in Table 7: a 2×2 crossing of objective {predictive, reconstructive} with architecture {CNN, CNN+ViT} at matched $d = 64$, with the auxiliary heads matched across all four cells, evaluated under the same predictor and probes; together with a predictive variant from which the wake head is removed. The decisive question is whether the predictive objective still closes the wake observables better than the reconstructive one once the architecture and the wake supervision are held in common. [PENDING: run the auxiliary-matched 2×2 and the no-wake-head JEPA; populate Table 7; then state, per the evidence, whether the claim is “predictive objective improves closure”, “predictive encoder family improves closure”, or “the wake supervision improves closure”.]

4.6 Physical-space interpretation

Forward closure is a statement about numbers; the encounter is a statement about vortices. Decoded and reconstructed vorticity fields on held-out encounters (Figures 5–7) connect the two, with one methodological caution and one new metric.

The caution is that field-space similarity scores are misleading here. On a held-out interpolation encounter the reconstructive autoencoder attains a high structural similarity yet a peak

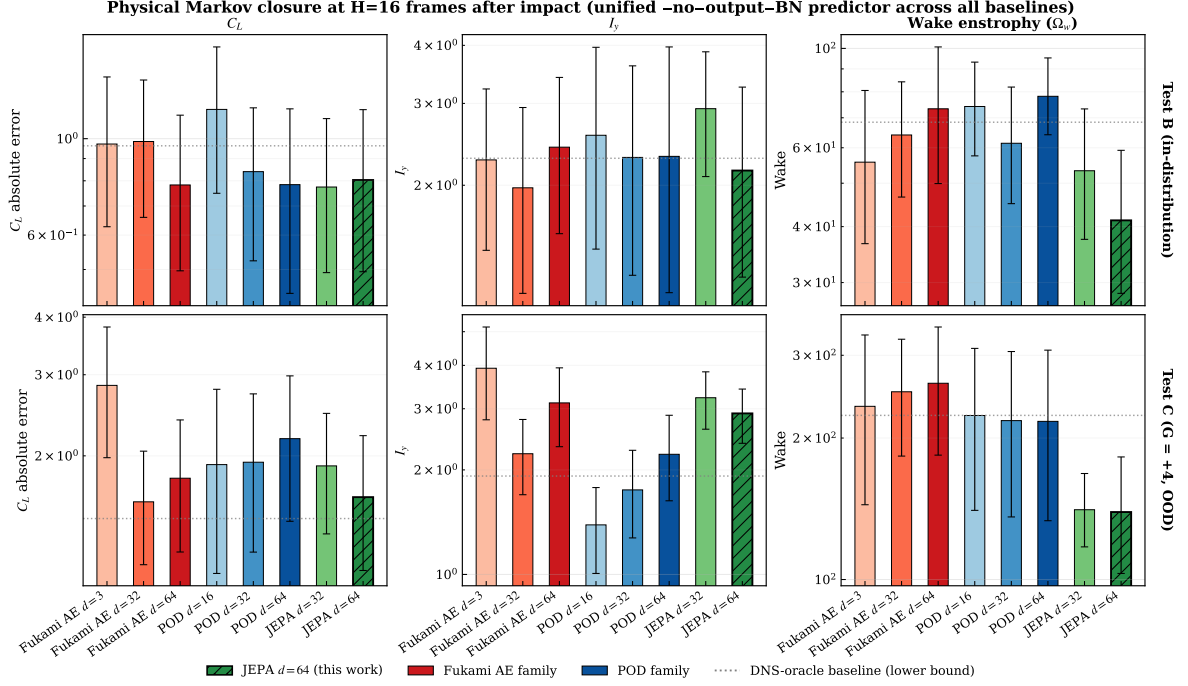


Figure 4: Held-out forward closure at $H = 16$ for the force, impulse, and wake observables across all baselines on test_b (in-distribution, top) and test_c ($|G| = 4$ extrapolation, bottom). Error bars are bootstrap 95% confidence intervals over $n = 42$ and $n = 24$ encounters. The wake panel shows the clearest separation between the predictive latent and the reconstructive and linear families.

reconstructed vorticity magnitude of only 0.36 against a target peak near 9.7: it has collapsed to a near-uniform mean field and lost the vortex-impingement signature, and the high similarity merely reflects bulk agreement with a field that is itself mostly quiescent away from the impact region. The predictive latent localises the impingement at the right position and sign (peak magnitude 5.31) but its visualisation decoder is blurry by construction, and the linear basis is amplitude-accurate (peak 4.10) since its objective is amplitude-optimal.

The new metric resolves the caution. An unbalanced optimal-transport distance between the reconstructed and the simulation fields, computed for the positive and negative vorticity separately and summed [19], penalises the collapsed reconstruction correctly (a large transport cost to move a quiescent field onto the true LEV) and rewards the correctly localised one, where structural similarity does the opposite. Recomputing the reconstruction comparison under this transport metric is the honest replacement for the similarity score. **[PENDING: recompute the reconstruction comparison under the OT field distance.]**

Finally, the wake observables can be tied to the structures that carry the load. A scale decomposition that separates the large-scale vortical structures responsible for the lift peaks from the fine-scale turbulence [15, 14], applied to the simulation field and to each family’s reconstruction, would show which scales the predictive latent retains and the reconstructive one smooths away, narrated through the staged encounter of §2.1: the gust-induced leading-edge vorticity flux, the LEV roll-up, the trailing-edge shedding at peak load, and the recovery. This is where the wake-entropy and circulation advantage of §4.1 becomes a statement about the leading-edge vortex and the shear layer rather than about a scalar. **[PENDING: add the scale-decomposition interpretation of the wake-observable advantage.]**

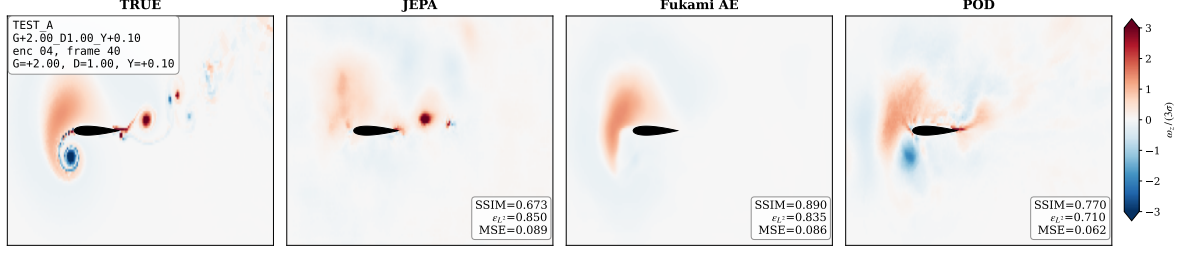


Figure 5: Reconstruction on a held-out test_a encounter (trained gust case, held-out encounter): simulation, predictive-latent decode, reconstructive autoencoder, and POD. The predictive decoder is blurry by design but localises the structures; field-space similarity scores are reframed by the optimal-transport metric of §4.6.

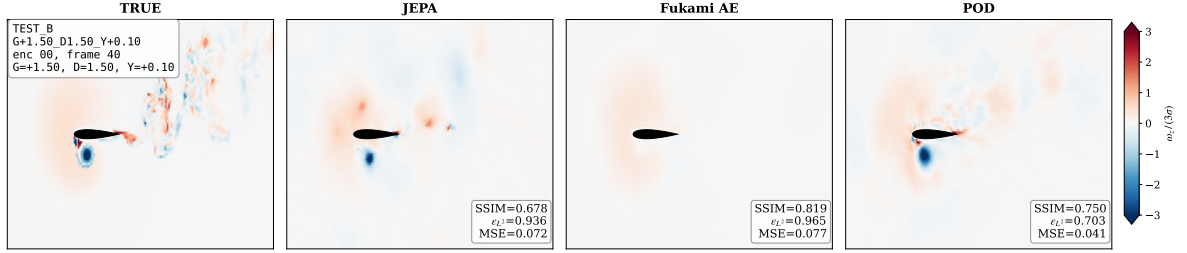


Figure 6: Reconstruction on a held-out test_b interpolation encounter (unseen interior (G, D, Y)). The reconstructive autoencoder has collapsed to a near-uniform field at this unseen condition while the predictive latent retains the impingement geometry; see §4.6.

5 Discussion

5.1 What the predictive latent retains, and why

The central finding is about dynamic closure, not reconstruction. Under a shared predictor, parametric conditioning, and probe family, the predictive latent family keeps the forward evolution of six physical observables closer to the simulation than the reconstructive and linear families do, and the advantage is concentrated on the spatially distributed wake observables that integrate over the shedding region. The mechanism is geometric. A reconstruction objective fixes the latent only up to a diffeomorphism [19, 10], so the decoder is free to place codes anywhere it finds convenient and nothing constrains the latent geometry to be predictable in time; under autoregressive rollout the reconstructive latent therefore leaves its own training manifold, and a probe queried on that drifted state fails for reasons unrelated to the predictor’s one-step accuracy. The predictive objective, regularised against collapse, instead organises the latent so that the encounter is a closed cycle that survives rollout and so that latent distance tracks the physical transport geometry of the flow, which is why iterating the predictor does not leave the manifold. The wake observables are the sharpest discriminator precisely because they can change while the scalar forces do not: a representation can carry the lift signature of an encounter without carrying the vorticity redistribution that produces it, and the reconstructive latent does exactly that, recovering the impact lift poorly from its own oracle while the predictive latent recovers it well.

5.2 What the comparison does and does not establish

We are deliberate about the scope of the claim. The predictive encoder here differs from the reconstructive baseline in three respects at once: the training objective, the CNN+ViT architecture, and the wake supervision. The result as it stands is therefore a statement about the

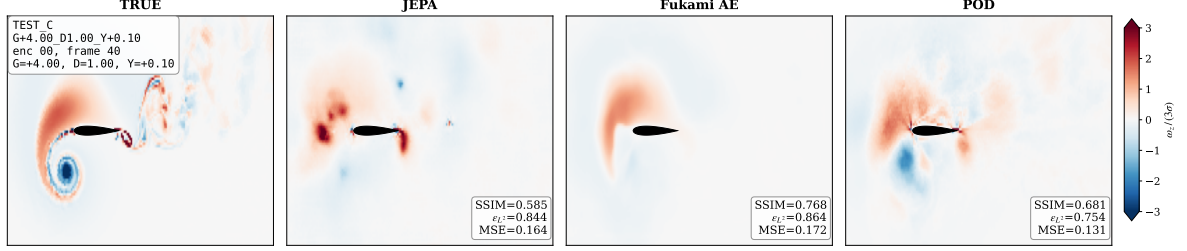


Figure 7: Reconstruction on a held-out test_c extrapolation encounter at $|G| = 4$, the three-dimensional regime of §2.1. All families degrade; the comparison is reported as the observability boundary rather than as a generalisation claim.

predictive latent *family* as configured. The conditioning-only floor removes the shared parametric conditioning as an explanation, but the architecture and, most pointedly, the wake supervision remain confounded with the objective, and the largest part of the advantage is on the very wake observables the predictive encoder was supervised on. Attributing the gain to the predictive objective in isolation requires the matched-architecture, matched-auxiliary controls and the no-wake-head variant of §4.5; until those are in hand the safe statement is “the predictive latent family improves forward closure”, not “the predictive objective alone does”. The linear basis, for its part, remains genuinely useful: it is the most competitive family on the integrated impulse and is amplitude-accurate in reconstruction, so the result is not that POD is obsolete but that it does not produce a latent whose rollout closes the wake observables.

5.3 Limitations

Three limitations bound the claims. First, the wall-normal offset Y is weakly resolved because the training mass concentrates near $Y \approx 0$; this is a data limitation, and Y -conditional statements should be read with it in mind. Second, the $|G| = 4$ extrapolation degrades for a physical reason, not merely a statistical one: the encoder observes a single mid-plane vorticity slice, and at $|G| = 4$ the true dynamics become three-dimensional [6], so test_c is best read as the observability boundary of a mid-plane representation rather than as a parametric-generalisation claim. Third, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss; a reader expecting sharp reconstructions from a predictive latent will be disappointed, which is the explicit trade the objective makes. To these we add the objective-isolation caveat of §5.2 as a fourth.

5.4 Pathway to model-based control, and a pilot that does not yet close

The predictor is, by construction, the latent dynamics function a model-based controller plans against, and three properties of the architecture support that use: closure holds out to the planning horizon relevant to the impact-instant load, the latent is two to three orders of magnitude cheaper to evolve than the full field, and the rollout stays on the training manifold, which is the property that determines whether long-horizon plans remain meaningful since a planner queries the predictor on states reached by its own rollout. Because the encoder is unconditional, the latent itself is the observable a controller has access to, and the gust parameters enter only the predictor; replacing the parametric conditioning with an actuation channel is a change of input rather than of architecture, and the phase-amplitude reading of §4.4 connects the predictor directly to the sensitivity-function control designed for these flows [5].

We also report, as a limitation rather than a result, a closed-loop pilot that chains a wall-pressure state estimator, the predictor rollout, and an observable probe. A temporal-convolutional estimator recovers the latent from a pre-impact pressure window well (held-out $R^2 \approx 0.87$ at eight sensors), but the assembled loop does not meet the tolerances set in advance: at the design

horizon only a small fraction of held-out encounters fall within the C_L and I_y tolerance bands, and none of the pre-registered gates are met. The bottleneck is the rollout rather than the estimator, since the within- tolerance fraction is essentially unchanged when the oracle latent is substituted for the estimated one. The honest reading is that the representation and the pressure observability are in place but the rollout needs reinforcement, through more frequent re-estimation from pressure during the rollout or a longer scheduled-sampling horizon, before a usable controller follows. The estimator, sensor-placement, and closed-loop details are deferred to Appendix C.

5.5 Outlook

The contribution is narrow by design: a controlled comparison on a single parametric flow, anchored to forward physical closure rather than to in-distribution reconstruction, with a drift mechanism that explains the gap and a set of controls that bound its interpretation. The immediate next step is to run those controls and the horizon, topology, transport, and phase-amplitude analyses set out above, which together would either promote the claim to the predictive objective in isolation or sharpen it to the predictive family. Beyond that, applying the same matched-predictor closure protocol to other parametric flow datasets would test how general the predictive-versus-reconstructive trade reported here actually is.

6 Conclusions

We compared a predictive latent (JEPA), a reconstructive observable-augmented autoencoder, and a proper orthogonal decomposition basis as reduced states for parametric vortex-gust airfoil interactions at $Re = 5000$, under a protocol in which a single autoregressive predictor, conditioning, and probe family were held fixed so that only the encoder varied, and judged each by forward physical closure rather than by reconstruction. On held-out encounters the predictive latent kept the wake observables closest to the simulation, by a factor of two to three over the other families on wake enstrophy, while the linear basis remained most competitive on the integrated impulse. The mechanism is rollout drift: the reconstructive latent leaves its training manifold by an order of magnitude over the forecast horizon, so that the wake information hidden from a reconstruction objective becomes recoverable from a predictive one, whereas the predictive and linear latents stay in distribution. Because the predictive encoder also differs in architecture and auxiliary supervision, we attribute the result to the predictive latent family and set out the matched controls that would isolate the objective. The predictor is the latent dynamics function of model-based control; a closed-loop pilot does not yet meet its tolerances, with the rollout rather than the state estimator as the bottleneck, and is reported as the boundary of what the present representation supports.

References

- [1] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Balas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.
- [2] M. Assran et al. V-JEPA 2: self-supervised video models. *arXiv preprint arXiv:2506.09985*, 2025.
- [3] R. Balestrierio and Y. LeCun. LeJEPA: provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

- [4] A. Bardes, J. Ponce, and Y. LeCun. VICReg: variance-invariance-covariance regularization for self-supervised learning. In *ICLR*, 2022.
- [5] K. Fukami, H. Nakao, and K. Taira. Data-driven transient lift attenuation for extreme vortex gust–airfoil interactions. *J. Fluid Mech.*, 992:A17, 2024.
- [6] K. Fukami, L. Smith, and K. Taira. Extreme vortex-gust airfoil interactions at Reynolds number 5000. *Phys. Rev. Fluids*, 10:084703, 2025.
- [7] K. Fukami and K. Taira. Grasping extreme aerodynamics on a low-dimensional manifold. *Nat. Commun.*, 14:6480, 2023.
- [8] K. Fukami and K. Taira. Observable-augmented manifold learning for multi-source turbulent flow data. *J. Fluid Mech.*, 1010:R4, 2025.
- [9] A. R. Jones, O. Cetiner, and M. J. Smith. Physics and modeling of large flow disturbances: discrete gust encounters for modern air vehicles. *Annu. Rev. Fluid Mech.*, 54:469–493, 2022.
- [10] D. Kelshaw and L. Magri. Proper latent decomposition. *arXiv preprint arXiv:2412.00785*, 2024.
- [11] Y. LeCun. A path towards autonomous machine intelligence. Technical report, OpenReview, 2022. Version 0.9.2.
- [12] P.-A. Maes, Q. Garrido, M. Assran, A. Bardes, M. Rabbat, Y. LeCun, and N. Ballas. LeWM: latent world models for self-supervised video prediction. *arXiv preprint arXiv:2603.19312*, 2026.
- [13] A. Mohamed et al. Opportunities and challenges of gust-aware flight. *Prog. Aerosp. Sci.*, 2023. verify volume/pages.
- [14] Y. Motoori and S. Goto. Generation mechanism of a hierarchy of vortices in a turbulent boundary layer. *J. Fluid Mech.*, 865:1085–1109, 2019.
- [15] H. Odaka, B. Lopez-Doriga, and K. Taira. Vortical similarities across laminar and turbulent extreme gust encounters. *J. Fluid Mech.*, 1031:R3, 2026.
- [16] L. Smith, K. Fukami, G. Sedky, A. Jones, and K. Taira. A cyclic perspective on transient gust encounters through the lens of persistent homology. *J. Fluid Mech.*, 980:A18, 2024.
- [17] V. Sobal, W. Zhang, K. Cho, R. Balestrieri, T. G. J. Rudner, and Y. LeCun. Learning from reward-free offline data: a case for planning with latent dynamics models. *arXiv preprint arXiv:2502.14819*, 2025.
- [18] A. Solera-Rico, C. Sanmiguel Vila, M. Gómez-López, Y. Wang, A. Almashjary, S. T. M. Dawson, and R. Vinuesa. β -variational autoencoders and transformers for reduced-order modelling of fluid flows. *Nat. Commun.*, 15:1361, 2024.
- [19] J. Tran, C.-A. Yeh, and K. Taira. Using optimal transport aligned latent embeddings for separated flow analysis. *J. Fluid Mech.*, 1027:A24, 2026.